

Bufotalin 逆合成审计报告

基于 Tetrahedron 2025 本地全文/PDF 的 source-detail 结构识别与 harness 编译结果。本报告区分文献确定路径、ChemEnzy 探索路径和 verifier 拒绝的 raw solved 路径。

文献链

15 steps

one-step rows

15

compiler

executable_ready

ChemEnzy verifier

rejected

核心结论

- 文献路径完整接到 compound 11, compound 11 被作为 androstenedione chiral-pool anchor 处理；外部 stock closure 仍需 stock resolver。
- source-detail resolver/compiled downstream 已产生 15 条 executable one-step rows, 可被 literature_template_plugin 消费。
- ChemEnzy raw_solved=true 被 deterministic verifier 拒绝, 原因包括 advanced_same_scaffold_terminal、large_atom_jump 和 no_verifier_accepted_stock_closed_route。
- 推荐策略是 hybrid: 文献链作为高权重 baseline, 同时保留 ChemEnzy 对上游和替代路线的探索能力。

主文献: doi:10.1016/j.tet.2025.134610 | Construction of advanced intermediate sharing C14-beta-OH for the synthesis of bufotalin
生成时间: 2026-06-06T17:50:16.346852+00:00

文献逆合成路线总览



- 后期 33 -> bufotalin: HF-pyridine 脱保护, 产率 93%。
- C17 2-pyrone 单元在 14 -> 22 步接入; harness 默认只沿 steroid 主骨架作为 child target 继续扩展。
- C14/C16 氧化态在 20、14、22、30、31、32、33 阶段完成后期调整。
- 11 -> 24 -> ... -> 20 对应 steroid semisynthesis / chiral-pool 方向, 不从头构建四环甾体骨架。

说明: 该路线是 source-detail 文献 baseline, 不等价于强制唯一路线; ChemEnzy 探索仍保留, 但必须通过 verifier 和 stock closure。

source-detail 逆合成步骤表 (1/2)

1. 33 -> bufotalin: global silyl deprotection

HF-pyridine; THF; room temperature; 160 h

yield: 93% | source: doi:10.1016/j.tet.2025.134610

Bufotalin is obtained from 33 with HF-pyridine in THF at room temperature.

2. 32 -> 33: C3 acetylation

Ac2O; pyridine; room temperature; 20 h

yield: 90% | source: doi:10.1016/j.tet.2025.134610

Compound 33 is obtained from 32 with Ac2O in pyridine at room temperature.

3. 31 -> 32: ketone reduction

NaBH4; MeOH; THF; 0 C; 2 h

yield: 90% | source: doi:10.1016/j.tet.2025.134610

Compound 32 is obtained from 31 with NaBH4 in MeOH/THF at 0 C.

4. 30 -> 31: Lewis-acid rearrangement

TMSOTf; 2,6-lutidine; DCM; -78 C to room temperature; 1 h

yield: 68% | source: doi:10.1016/j.tet.2025.134610

Compound 31 is obtained from 30 with TMSOTf and 2,6-lutidine in DCM.

5. 22 -> 30: epoxidation

m-CPBA; Na2CO3; DCM; -78 C to -20 C; 10 h

yield: 70% | source: doi:10.1016/j.tet.2025.134610

Compound 30 is obtained from 22 using m-CPBA and Na2CO3 in DCM.

6. 14 -> 22: C17 2-pyrone coupling

compound 21; Pd(PPh3)4; CuI; LiCl; DMSO; THF; 60 C; 12 h

yield: 62% | source: doi:10.1016/j.tet.2025.134610

Compound 22 is obtained from 14 and compound 21 under Pd/Cu/LiCl coupling conditions.

7. 20 -> 14: vinyl iodide formation

Pd/C, H2; hydrazine monohydrate; triethylamine; I2; THF/water; EtOH; THF; room temperature; 50 C; room temperature; 12 h; 6 h; 1 h

yield: 73% over two steps | source: doi:10.1016/j.tet.2025.134610

Compound 14 is obtained from 20 by hydrogenation followed by hydrazone formation and iodination.

8. 19 -> 20: C14 allylic oxidation

SeO2; formic acid; dioxane/water 4:1; 125 C; 16 h

yield: 52% | source: doi:10.1016/j.tet.2025.134610

Compound 20 is obtained from 19 by SeO2 allylic oxidation.

source-detail 逆合成步骤表 (2/2)

9. 28 -> 19: C3 TBS protection

TBSCl; imidazole; DMF; 70 C; 16 h

yield: 74% | source: doi:10.1016/j.tet.2025.134610

Compound 19 is obtained from 28 by C3 TBS protection.

10. 27 -> 28: C17 deketalization

p-TsOH monohydrate; acetone; 60 C; 1.5 h

yield: 56% | source: doi:10.1016/j.tet.2025.134610

Compound 28 is obtained from 27 by deketalization.

11. 26 -> 27: elimination

t-BuOK; DMSO; 70 C; 16 h

yield: 73% | source: doi:10.1016/j.tet.2025.134610

Compound 27 is obtained from 26 by elimination to the D-ring olefin.

12. 23 -> 26: C16 bromination

pyridinium perbromide; THF; 0 C to room temperature; 2 h

yield: 87% | source: doi:10.1016/j.tet.2025.134610

Compound 26 is obtained from 23 by C16 bromination.

13. 25 -> 23: stereoselective reduction

K-selectride; THF; -20 C to -5 C; 10 min then warm

yield: 72% | source: doi:10.1016/j.tet.2025.134610

Compound 23 is obtained from 25 by C3 ketone reduction.

14. 24 -> 25: stereoselective hydrogenation

Pd/C; H₂; 4-methylpyridine; room temperature; 23 h

yield: 92% | source: doi:10.1016/j.tet.2025.134610

Compound 25 is obtained from 24 by stereoselective hydrogenation.

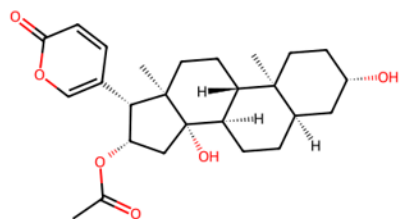
15. 11 -> 24: C17 ketalization

p-TsOH monohydrate; ethylene glycol; ethylene glycol; room temperature; 2 h

yield: 93% | source: doi:10.1016/j.tet.2025.134610

Compound 24 is obtained from androstenedione 11 by C17 ketalization.

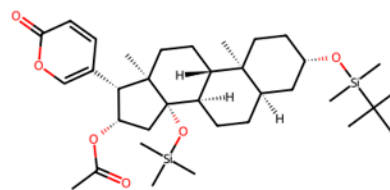
结构与公式核验 (1/2)



bufotalin C₂₆H₃₆O₆

bufotalinC₂₆H₃₆O₆

exact MW 444.2512 | heavy atoms 32

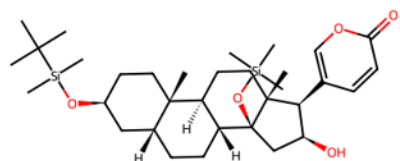


33 C₃₅H₅₈O₆Si₂

33

C₃₅H₅₈O₆Si₂

exact MW 630.3772 | heavy atoms 43

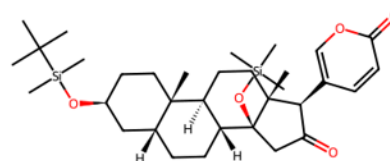


32 C₃₃H₅₆O₅Si₂

32

C₃₃H₅₆O₅Si₂

exact MW 588.3666 | heavy atoms 40

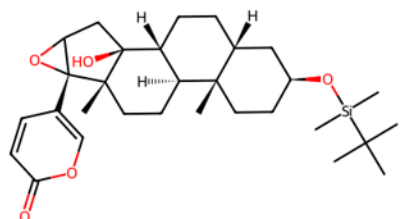


31 C₃₃H₅₄O₅Si₂

31

C₃₃H₅₄O₅Si₂

exact MW 586.351 | heavy atoms 40

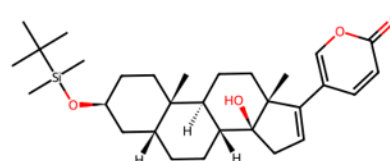


30 C₃₀H₄₆O₅Si

30

C₃₀H₄₆O₅Si

exact MW 514.3115 | heavy atoms 36

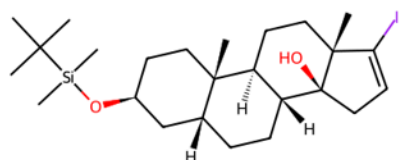


22 C₃₀H₄₆O₄Si

22

C₃₀H₄₆O₄Si

exact MW 498.3165 | heavy atoms 35

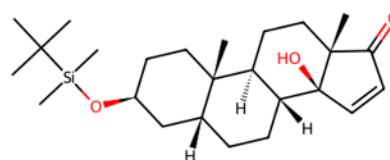


14 C₂₅H₄₃O₂Si

14

C₂₅H₄₃O₂Si

exact MW 530.2077 | heavy atoms 29



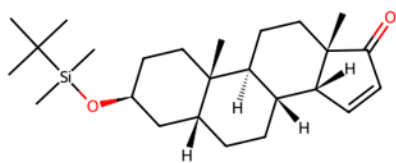
20 C₂₅H₄₂O₃Si

20

C₂₅H₄₂O₃Si

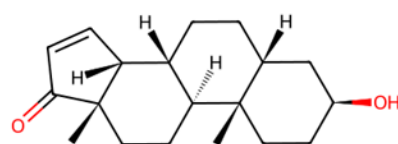
exact MW 418.2903 | heavy atoms 29

结构与公式核验 (2/2)



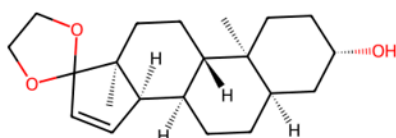
19 C₂₅H₄₂O₂Si

19 C₂₅H₄₂O₂Si
exact MW 402.2954 | heavy atoms 28



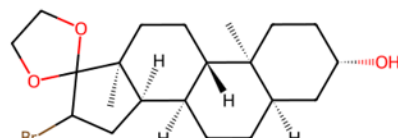
28 C₁₉H₂₈O₂

28 C₁₉H₂₈O₂
exact MW 288.2089 | heavy atoms 21



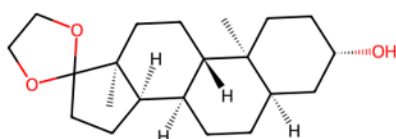
27 C₂₁H₃₂O₃

27 C₂₁H₃₂O₃
exact MW 332.2351 | heavy atoms 24



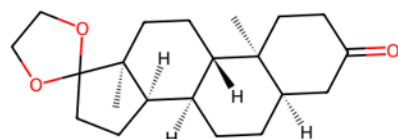
26 C₂₁H₃₃BrO₃

26 C₂₁H₃₃BrO₃
exact MW 412.1613 | heavy atoms 25



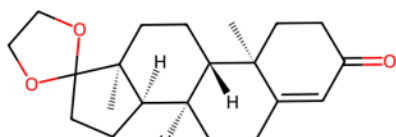
23 C₂₁H₃₄O₃

23 C₂₁H₃₄O₃
exact MW 334.2508 | heavy atoms 24



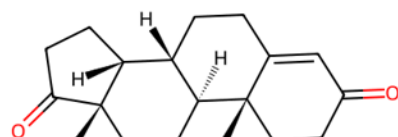
25 C₂₁H₃₂O₃

25 C₂₁H₃₂O₃
exact MW 332.2351 | heavy atoms 24



24 C₂₁H₃₀O₃

24 C₂₁H₃₀O₃
exact MW 330.2195 | heavy atoms 24



11 C₁₉H₂₆O₂

11 C₁₉H₂₆O₂
exact MW 286.1933 | heavy atoms 21

Harness 编译与 ChemEnzy 对照

curator steps 15	source-detail rows 15	one-step rows 15
child targets 15	raw ChemEnzy routes 14	verifier accepted 0

- compiled_accepted=True, executable_status=executable_ready。
- literature_template_plugin enabled=True, max_added=15。
- source-detail chain probe accepted=True, terminal_reached=True, 确认 15 步链能从 bufotalin unroll 到 compound 11。
- ChemEnzy raw_solved=true 只说明 native core 返回 stock-closed routes ; deterministic verifier 拒绝全部 raw routes, 因此不能宣布 solved。

Verifier rejection reasons: advanced_same_scaffold_terminal, large_atom_jump, no_verifier_accepted_stock_closed_route

- guided ChemEnzy rerun 可以直接消费 15 条 one-step rows。
- route expansion 可按 source-detail child targets 逐步探索上游, 但不把文献路径强制设为唯一解。
- compound 11 的 commercial/stock closure 仍应由 stock resolver 或采购数据库独立确认。

证据与可复核产物

literature_path_high_weight_baseline: True
literature_path_not_mandatory_replacement: True
chemenzy_exploration_retained: True
raw_solved_not_equivalent_to_verified_solved: True
no_solved_claim: True
production_write_blocked: True

source_detail_chain_audit	/root/autodl-tmp/AutoPlanner/results/shared/bu fotalin_tet2025_pdf_fullroute_to_androstenedio ne_source_detail_20260607/source_detail_route_
source_detail_curator_records	/root/autodl-tmp/AutoPlanner/results/shared/bu fotalin_tet2025_pdf_fullroute_to_androstenedio ne_source_detail_20260607/source_detail_curato
compiled_downstream	/root/autodl-tmp/AutoPlanner/results/shared/bu fotalin_tet2025_pdf_fullroute_to_androstenedio ne_source_detail_20260607/compiled_downstream_
chemenzy_smoke_summary	/root/autodl-tmp/AutoPlanner/results/shared/bu fotalin_tet2025_fullroute_15row_smoke_rerun_20 260607/summary.json
chemenzy_verifier	/root/autodl-tmp/AutoPlanner/results/shared/bu fotalin_tet2025_fullroute_15row_smoke_rerun_20 260607/verifier.json
chain_tool_probe	/root/autodl-tmp/AutoPlanner/results/shared/bu fotalin_harness_chain_tool_probe_20260607_term inal_fixed/source_detail_route_chain_audit.js

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